



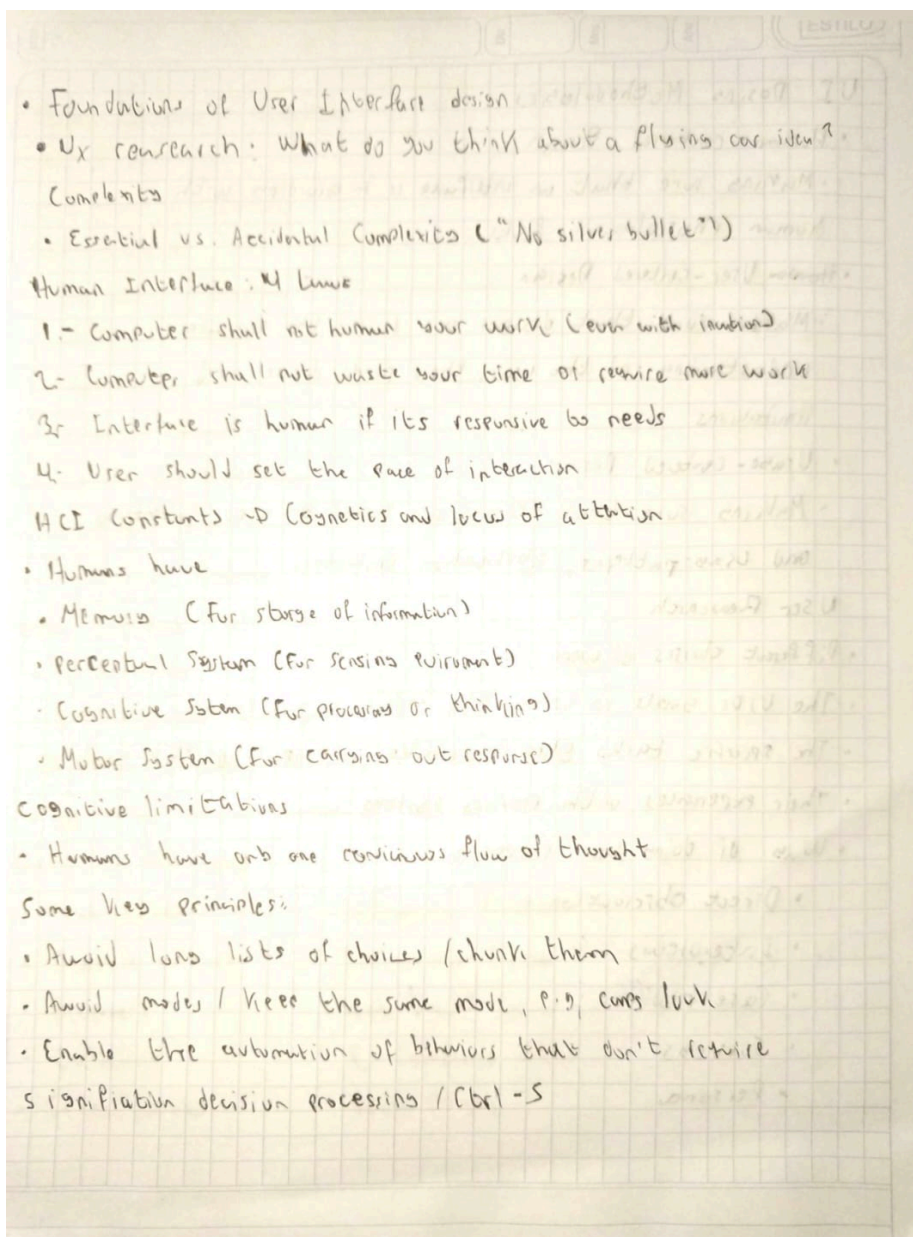
ESPE
UNIVERSIDAD DE LAS FUERZAS ARMADAS
INNOVACIÓN PARA LA EXCELENCIA

Advanced Web Development

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UI Design Methodologies

- Human-centered Design
 - Making sure that an interface is in accordance with human physiological facts
- Human User-centered Design
 - Making sure that designs are driven by accurate understanding of the user, their needs, desires, and limitations
- User-centered Design
 - Making sure that designs are based on user intentions and usage patterns, abstraction prototypes

User Research

- Different classes of users
- The user's goals in using the software you design
- The specific tasks they undertake in pursuit of those goals
- Their experiences with previous systems
- Ways of doing user research
 - Direct observation
 - Interviews
 - Case studies
 - Surveys
 - Personas

User Behavioral Patterns

- Self exploration
- Instant Gratification
- Satisficing
- Changes in Midstream
- Deferred choices
- Habituation
- Spatial Memory
- Streamlined Repetition
- Keyboard only
- Prospective Memory

User Interface Metaphors

- A desktop
- A control panel metaphor
- A store or shopping metaphor
- An art room metaphor

Interface Idioms

- | | |
|-------------------|----------------------|
| • Text Editors | Information Graphics |
| • Graphic editors | Immersive Game |
| • Browsers | Web Social Sences |
| • Calendars | E-commerce Sites |
| • Spreadsheets | Web Pages |

Page Layout

- Being an Art doesn't imply a kind of rationale or design.
- Five elements
 - Visual Hierarchy: The most important content should stand out the most.
 - Visual Flow: Is how your eye moves across a screen or page.
 - Grouping and Alignment: Humans naturally want order.
 - Dynamic Displays: Unlike printed pages or posters.
 - Integration of these Elements

Patterns

- Visual Framework: Design each page to use the same basic layout, colors, and stylistic elements.
- Center Stage: Put the most important part of the UI into the largest subsection of the page or windows.
- Card Stack
 - Titled Sections
 - Diagonal Balance
 - Responsive Displacement
 - Liquid layout
 - Properties Sheet
 - Module Tabs
 - Accordion
 - Grid of Equals.

• Navigation

Four Questions

- Where am I now?
- Where do I want to go?
- How do I get there?
- Am I there yet?

Signpost

- Window title
- Logos
- Other branding devices (color schemes, tab lines, etc.)
- Tabs
- Selection indications

"You are here" patterns

- Sequence Map
- Breadcrumbs
- Annotated Scrollbar

Wayfinding

- Is what people (users) do as they find their way towards their goals
- Wayfinds and the structure of a system are directly dependent on each other
- "Wayshowing" is what GUI developers need to do - Facilitate wayfinding.

Actions and Commands

- Make the desired actions available
- Label them well
- Make them easy to find
- Support sequences of actions

Mechanisms

• Visible

- Buttons
- Menu Bars
- Pop-up menus
- Toolbars
- Links

• Invisible

- Single clicking
- Double clicking
- Keyboard actions
- Drag-and-drop

Patterns

- Hover Tools
- Action Panels
- Smart Menu items
- Preview
- Progress indicator

• Cancelability

• Command Histories

• Menus

• Intelli-sense

- A single gesture causing multiple actions

Showing Complex Data

Information Graphics

- Graphics

- Tables

- Trees

- Maps

- Considers the spatial relationships in the presentation to convey meaning

View activities

- User research

- We need to understand the users:

- Their attitudes

- Their priorities

- Their abilities and limitations

- System Analysis

- We need to understand the real-world

- Problems that the users need to solve

- We need to understand the available data

Shape and Appearance

- The "shape" of data presentation conveys meaning

- The "look" also conveys meaning

Getting Input from the User

Principles

- If you can, avoid asking the question at all
- Do usability testing
- Beware of a literal translation from the underlying

Programming model

Control Choice

- Factors to consider:
 - Available space
 - User sophistication with respect to
 - General computer usage
 - Domain knowledge
 - Expectations from other applications
 - Available technology

Controls

- For selection making a binary choice:
 - Single checkbox
 - Two radio buttons
- For N items, N is small
 - Spinner
 - N -item dropdown list

- N is large

- Custom browser

- N-item dropdown list with autocomplete

Patterns

- Text entry patterns

- Forgiving format

- Structured format

- Fill-in-the-blanks

- Input hints

- Input prompts

- Autocompletion

- Choice-related Patterns

- Dropdown chooser

- Illustrated choices

- List builder

- Other Patterns

- Good defaults

- Same-page error messages